SALESLENS ENTERPRISE BI PLATFORM • ACADEMIC DELIVERABLE DOC-03

Software Project Plan, WBS & CPM Schedule Management

Course: B.Tech CSE Semester III • Software Engineering & Project Management • Version 1.0

11	40.0 PD	7.8 Wks	2.67 Wks
WORK ACTIVITIES	TOTAL EFFORT	PROJECT DURATION	SCHEDULE BUFFER

1. Work Breakdown Structure (WBS Summary)

1.0 SalesLens Project: **1.1** Requirements Engineering & 9-View Selection (Week 1) • **1.2** Data Engineering & 14% Cleaning (Weeks 2-3) • **1.3** 9 Dashboard Views Build & RBAC (Weeks 3-5) • **1.4** Verification, BVA & DRE Testing (Weeks 6-7) • **1.5** Production Deployment & Conference Launch (Week 8).

2. Activity Precedence Matrix

ID	Activity Description	Effort (PD)	Resource	Predecessors	Duration (Days)
A	Requirements & 9-View Scope Signoff	3.0	BI Dev + Data Analyst	None	2.0
B	Architecture & DB Schema Design	3.0	BI Developer	A	3.0
C	14% Missing Region Code Cleaning Engine	8.0	BI Dev (4d) + Analyst (4d)	A	5.5
D	Automated Overnight ETL Pipeline Build	4.0	BI Developer	B, C	4.0
E	Views 1-3 Build (Pulse, Targets, Top SKUs)	9.0	BI Developer	D	9.0
F	Views 4-6 Build (Bottom SKUs, Region Matrix)	9.0	BI Developer	E	9.0
G	Views 7-9 Build (Categories, Channels, Audit)	9.0	BI Developer	F	9.0
H	RBAC Security & Filter Integration	2.0	BI Developer	G	2.0
I	System Testing, BVA & Decision Table QA	5.0	BI Dev (2d) + Analyst (3d)	H	3.5
J	User Acceptance Testing (UAT) Signoff	3.0	Analyst + Sales Director	I	2.0
K	Production Launch at Annual Conference	2.0	BI Dev + IT Admin	J	2.0

3. Critical Path Method (CPM) Forward & Backward Pass

Activity	Duration (D)	ES (Day)	EF (Day)	LS (Day)	LF (Day)	Total Float (Slack)	Critical Status
A	2.0	0.0	2.0	0.0	2.0	**0.0**	CRITICAL
B	3.0	2.0	5.0	4.5	7.5	**2.5**	NON-CRITICAL (SLACK 2.5D)
C	5.5	2.0	7.5	2.0	7.5	**0.0**	CRITICAL
D	4.0	7.5	11.5	7.5	11.5	**0.0**	CRITICAL
E	9.0	11.5	20.5	11.5	20.5	**0.0**	CRITICAL
F	9.0	20.5	29.5	20.5	29.5	**0.0**	CRITICAL
G	9.0	29.5	38.5	29.5	38.5	**0.0**	CRITICAL
H	2.0	38.5	40.5	38.5	40.5	**0.0**	CRITICAL
I	3.5	40.5	44.0	40.5	44.0	**0.0**	CRITICAL
J	2.0	44.0	46.0	44.0	46.0	**0.0**	CRITICAL
K	2.0	46.0	48.0	46.0	48.0	**0.0**	CRITICAL

Critical Path Determination:

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Critical Path = A → C → D → E → F → G → H → I → J → K
Optimized Duration: 39.0 working days = 7.8 weeks ≤ 8.0 weeks target (Includes 2 days reserve buffer)
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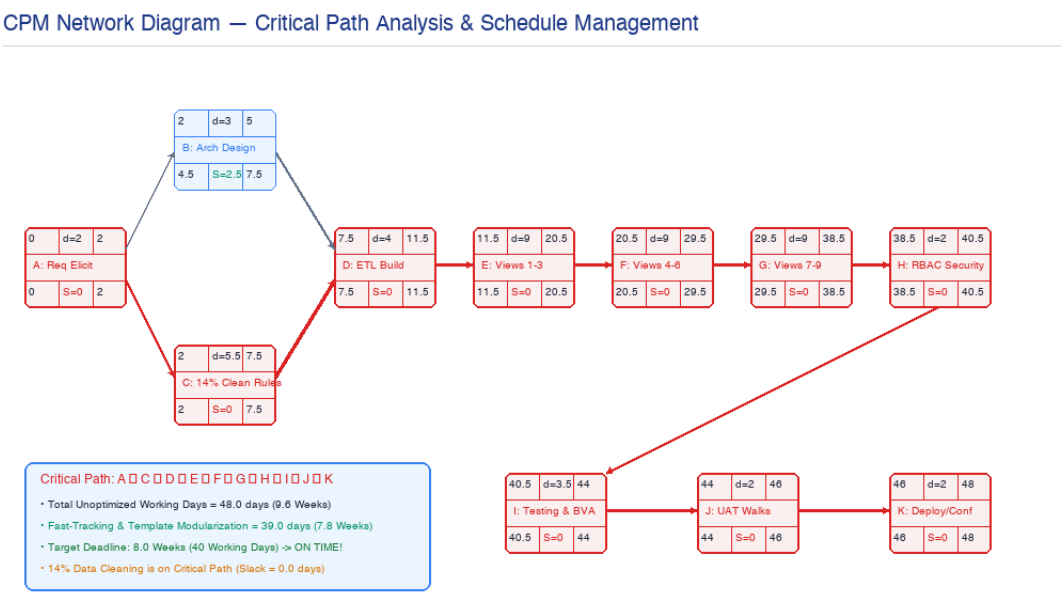


Figure 3.1: CPM Network Diagram showing Critical Path (Red) vs Non-Critical Design (Blue)

4. Master Gantt Chart Timeline & RACI Matrix

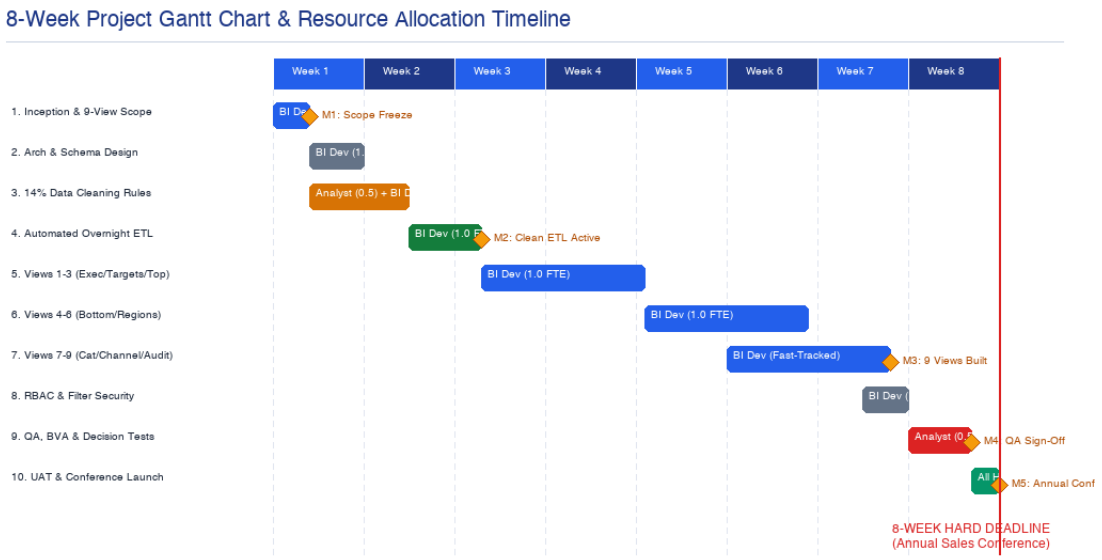


Figure 3.2: 8-Week Master Gantt Schedule across Inception, Cleaning, 9 Views, QA & Launch

Work Package	BI Developer (1.0 FTE)	Data Analyst (0.5 FTE)	Sales Director	IT Admin
Requirements & Scope	RESPONSIBLE	RESPONSIBLE	ACCOUNTABLE	CONSULTED
14% Data Cleaning Rules	RESPONSIBLE	RESPONSIBLE (SME)	INFORMED	CONSULTED
Overnight ETL Pipeline	RESPONSIBLE	CONSULTED	INFORMED	ACCOUNTABLE
9 Dashboard Views Build	RESPONSIBLE	CONSULTED	ACCOUNTABLE	INFORMED
BVA & QA Testing	RESPONSIBLE	RESPONSIBLE	INFORMED	CONSULTED
Conference Go-Live	RESPONSIBLE	RESPONSIBLE	ACCOUNTABLE	ACCOUNTABLE